

Rayyan Mohammed

☎ 8328004967 | ✉ studynow12463@gmail.com | 🔗 linkedin.com/in/rayyan-mohammed55 | 🐙 github.com/MDRayyan2006

SUMMARY

Computer Science undergraduate (CGPA: 8.95) with hands-on experience in AI systems, large language models (LLM), backend engineering, and full-stack development. Built production-grade multi-agent RAG systems, self-hosted MCP servers, and LangGraph-based hierarchical agent pipelines. Proficient in Python, LangChain, LangGraph, FAISS, HuggingFace Transformers, Groq, and REST API development. Seeking a software engineering internship or AI/ML engineering internship.

EDUCATION

Gayatri Vidya Parishad College of Engineering

B.Tech in Computer Science and Engineering, CGPA: 8.95/10

Visakhapatnam, AP

Expected Graduation: 2028

- **Coursework:** Data Structures and Algorithms, Database Management Systems, Operating Systems (*in progress*), Object-Oriented Programming, Computer Networks (*upcoming*)
- **Self-Study:** Machine Learning, Deep Learning, RAG Evaluation (RAGAS), Distributed Inference Systems, LangChain, HuggingFace Transformers, BM25, Model Context Protocol (MCP)

PROJECTS

HelioX — Multi-Agent RAG System 🔄

Python • LangChain • FAISS • Qdrant • BM25 • Elasticsearch • Redis • FastAPI

- Architected a hybrid RAG system combining vector search (FAISS/Qdrant), BM25 keyword retrieval, and entity-based indexing into a 4-stage pipeline: metadata filtering, entity matching, vector similarity search, and confidence-threshold fallback.
- Built a parallel worker agent layer with citation enforcement per chunk; designed an Adjudication Engine for semantic contradiction detection; enforced hallucination mitigation via confidence-threshold gating and evidence-bound generation; evaluated with RAGAS (Recall@K, MRR, Faithfulness Score).

Multi-Agent Collaboration System 🔄

Python • LangGraph • LangChain • Groq • Tavily • BeautifulSoup

- Designed a hierarchical multi-agent system using LangGraph StateGraph with a top-level supervisor routing tasks between a Research Team (Tavily web search, BeautifulSoup scraping) and a Writing Team (document outlining, writing, Python REPL chart generation).
- Integrated Groq high-speed LLM inference (Mixtral-8x7b-32768, LLaMA-3.3-70b) for agent execution; implemented specialized tool-calling agents with structured document I/O and code execution via Python REPL.

MCP Chat — Self-Hosted GitHub MCP Server and LLM Client 🔄

Python • Groq • MCP • GitHub API • LLM • prompt-toolkit

- Built a self-hosted MCP server and MCP client CLI connecting a large language model (LLM) via Groq inference to a Document MCP Server (in-memory retrieval and editing) and a GitHub MCP Server (10+ tools: repo management, issue tracking, README access).
- Implemented deterministic GitHub fallback logic in the MCP client to prevent LLM hallucinations on tool-call failures; engineered @doc retrieval and /command syntax with Tab auto-completion for reliable human-in-the-loop AI workflows.

BadgeVerse — Real-Time Chat App 🔄

React • Node.js • Express • MongoDB • Socket.IO • JWT • OAuth 2.0

- Full-stack real-time chat application with Socket.IO (sub-100ms delivery), JWT authentication, Google OAuth 2.0, MongoDB badge scoring system, and 15+ RESTful API endpoints with a responsive Tailwind CSS front-end.

PDF and Image Analyzer 🔄

Python • PyMuPDF • OpenCV • Pillow

- Extracted structured text, images, and metadata from 100+ page PDFs using PyMuPDF C bindings (approx. 3x faster than PyPDF2); built an OpenCV and Pillow preprocessing pipeline for OCR including denoising, contrast normalization, and deskewing.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, Java (basic)

Web Development: React, Node.js, Express, FastAPI, Tailwind CSS, Socket.IO, REST API, HTML, CSS

AI and ML: Large Language Models (LLM), RAG, RAG Pipelines, Multi-Agent Systems, LangChain, LangGraph, FAISS, HuggingFace Transformers, BM25, Qdrant, Prompt Engineering, NLP, RAGAS, Groq, Model Context Protocol (MCP), Distributed Inference, Agent Tool Calling

Databases: PostgreSQL, MongoDB, SQLite, Redis, Qdrant, Elasticsearch

DevOps and Tools: Docker, Kubernetes, Git, GitHub, GitHub API, GitHub Actions, Google Cloud Platform, Linux

Concepts: Data Structures and Algorithms, System Design, OOP, RESTful APIs, JWT, OAuth 2.0, SDLC, MCP Client-Server Architecture

Mathematics: Linear Algebra, Probability and Statistics, Calculus, Linear Regression, Gradient Descent

ACHIEVEMENTS AND ACTIVITIES

- **Smart India Hackathon (SIH)** — National level; built and demonstrated a working prototype within a 36-hour sprint. 2025
- **Quantum Valley Hackathon** — Competed against 100+ teams; developed an AI-assisted solution end-to-end. 2025
- Solved **100+ problems** on LeetCode covering arrays, linked lists, trees, graphs, dynamic programming, and sliding window.
- Independently designed and shipped 5 full-stack and AI/ML software engineering projects with no team or external guidance.